DIODES PCIBUDDY UTILITY: USER MANUAL

INTRODUCTION

Diodes builds a utility called PCIBUDDY, providing different modes with different functions for developers：

Function \ Mode	In-Band Mode	Out-Band Mode	Standalone Mode
Configuration Read / Write	O	O	
Read / Write / Save / Load for SPI-Flash and EEPROM	O	O	
RX Phy Eye	O	O	
LTSSM Status		O	
Error Counter	O	O	
Misc. Function	O	O	
Edit / Create *.bin/*.txt file for SPI-Flash and EEPROM			O

Note：

1) In-Band mode and Out-Band mode need Diodes Gen3 Switch products is plugged
2) Standalone mode need Diodes Gen3 Switch products is not plugged (PCI-BUDDY would check whether Diodes Gen3 Switch exist or not, if not, it will enter Standalone mode)
3) In-Band mode need to run under Windows 64-bit OS environment
4) Out-Band mode needs extra I2C tool (Aardvark host adapter) to connect to I2C pin of Diodes Gen3 Switch product. As the register is accessed via I2C, it is allowed to be used in a non-Windows OS environment

This document would introduces how to use this utility more detailed in the following chapter

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UTILITY REVISION

DATE	VERSION	DESCRIPTION
07/16/2020	0.1	Beta release of PCIPSGUI utility.
02/25/2021	0.1a	V0.1A release of PCIBUDDYII utility 1) Remove in-band & out-band mode, because the two modes has already merged into one mode 2) Replace interface screenshot
09/22/2021	0.1b	V0.1b 1) add In-Band mode & Out-Band mode environment and setting diagram for reference 2) Update content by the latest PCIBUDDY version (V5.01)
10/01/2021	0.1c	V0.1c 1) Update Tool Package content by V5.02 2) Modify wording
12/01/2021	0.1d	V0.1d 1) Modify Port-Lane Mapping chart in Phy-Eye chapter for all Gen3 Packet Switch
01/06/2022	0.1e	V0.1e 1) Modify User Manual based on V6.03 2) Add Standalone mode
06/07/2022	0.1f	V0.1f 1) Modify Gen3 Packet Switch Port-Lane Mapping chart in chapter 4.4
11/18/2022	0.1g	V0.1g 1) Modified description in Chapter 4.4 Phy Eye Tab
08/07/2023	0.1h	Update Tool Package content by V6.07 1) Add new feature “Save Subtree” in chapter 4.2
02/29/2024	0.1i	Update Tool Package content by V6.08 1) Fix silicon cannot find by In-band mode in Trouble Shooting 5.2

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1 PCIBUDDY Tool Package (V6.07)

aardvark.dll
PCIBuddy_II.exe
PI7C9X3G808-606GP.pdf
PI7C9X3G808GP_CDLEP.dat
PI7C9X3G808GP_ CDVEP.dat
PI7C9X3G808GP-606GP_TRANSPARENT.dat
PI7C9X3G816GP.pdf
PI7C9X3G816GP_ CDLEP.dat
PI7C9X3G816GP_ CDVEP.dat
PI7C9X3G816GP_TRANSPARENT.dat
PI7C9X3G1224GP.pdf
PI7C9X3G1224GP_ CDLEP.dat
PI7C9X3G1224GP_ CDVEP.dat
PI7C9X3G1224GP_TRANSPARENT.dat
PI7C9X3G1632GP.pdf
PI7C9X3G1632GP_TRANSPARENT.dat
PI7C9X3G1632GP_CDLEP.dat
PI7C9X3G1632GP_CDVEP.dat
PI7C9X3G1632GPQ.pdf
PI7C9X3G1632GPQ_TRANSPARENT.dat
PI7C9X3G1632GPQ_CDLEP.dat
PI7C9X3G1632GPQ_CDVEP.dat

PscDevice.dat
Pscpci.dll

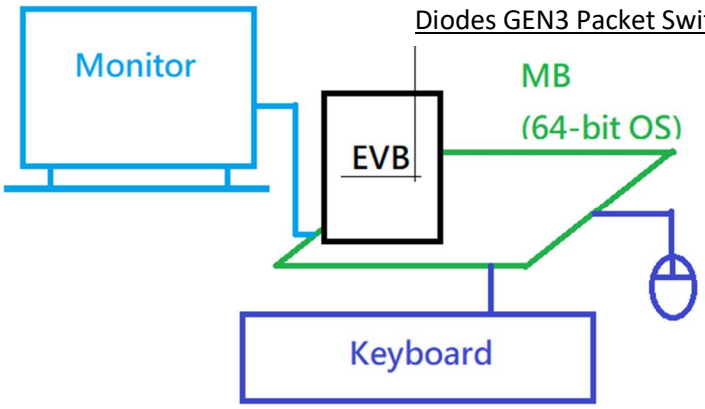
DUT (Diodes GEN3 Packet Switch EVB or Customer board with Diodes GEN3 switch IC)
Aardvark Host Adapter

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2 Setup

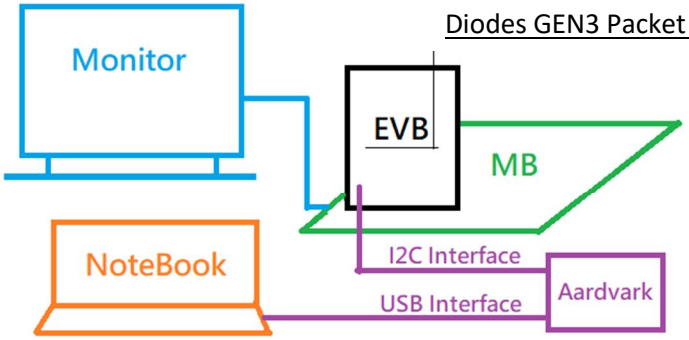
PCIBUDDY supports Microsoft Windows 64-bits OS, and targets NET framework version 4.5.2 or later. Aardvark I2C to USB dongle from Total Phase Inc. is used as the standard device for out-band process.
Please unzip *.rar first and then install TotalPhaseUSB-v3.0.0 (for Win10) or TotalPhaseUSB-v2.16 (for Win7).

- In-Band Mode setup and connection diagram：(for In-Band mode, user needs to run PCIBUDDY_II.exe as administrator)



In-Band mode environment and settting：
1) Make sure OS is 64-bit
2) Plug EVB into MB slot
3) Enter Win 64-bit system
4) Right click PCIBUDDY execute file and choose "Run as administrator" to enter In-Band mode

- Out-Band Mode setup and connection diagram：



Out-Band mode environment and setting：
1) Make sure Note Book OS is 64-bit
2) Plug EVB into MB slot
3) Turn on MB Power (provide power and clock to EEPROM)
4) Connect EVB I2C pin header and NB via Aardvark I2C Tool
5) double clock PCIBUDDY execute file to enter out-band mode on Note Book

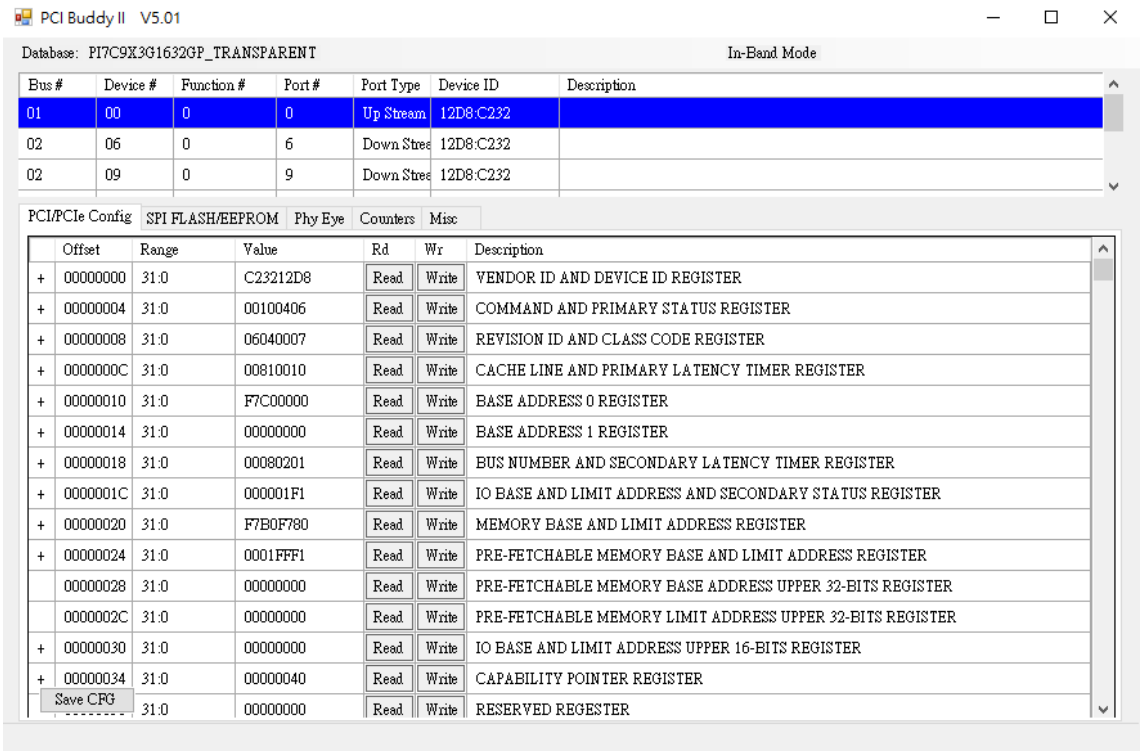
- Standalone Mode：PCIBUDDY would check whether Diodes Gen3 Switch exist or not, if not, it will enter Standalone mode. (No specific environment required)

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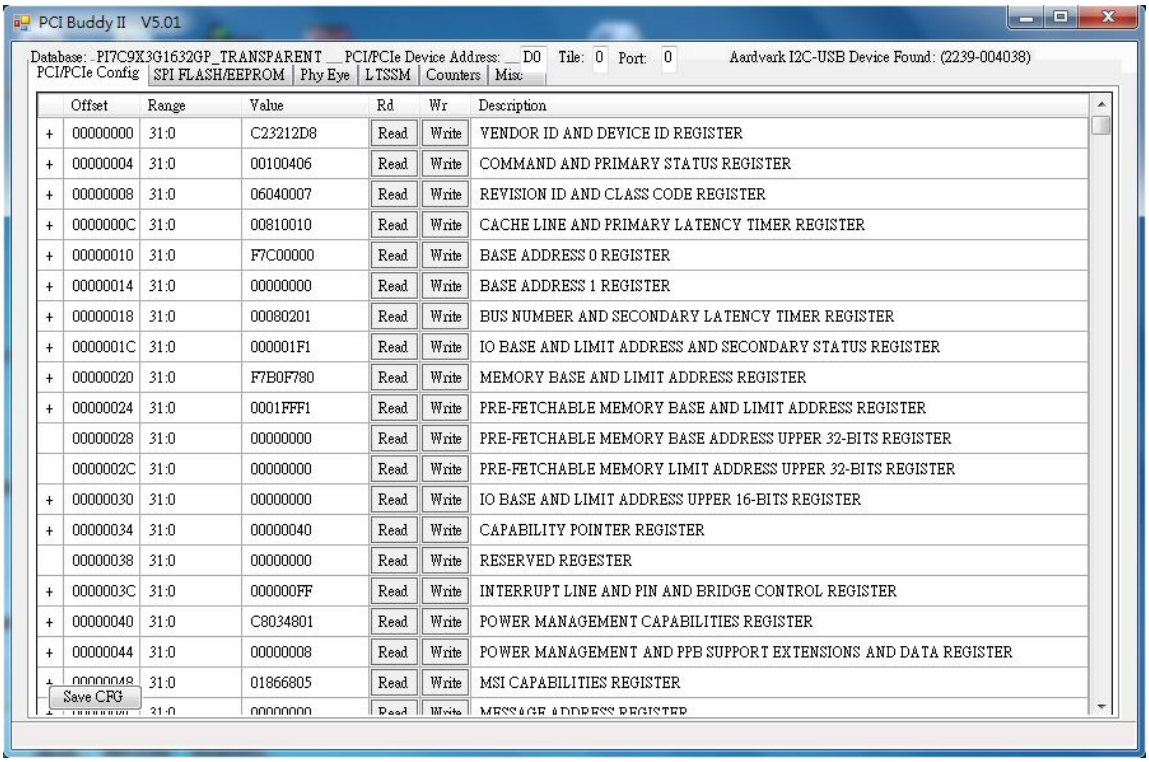
3 Run

Insert GEN3 EVB, run PCIBUDDY_II.exe through in-band or out-band will get different operation windows as bellow：

In-Band Mode：

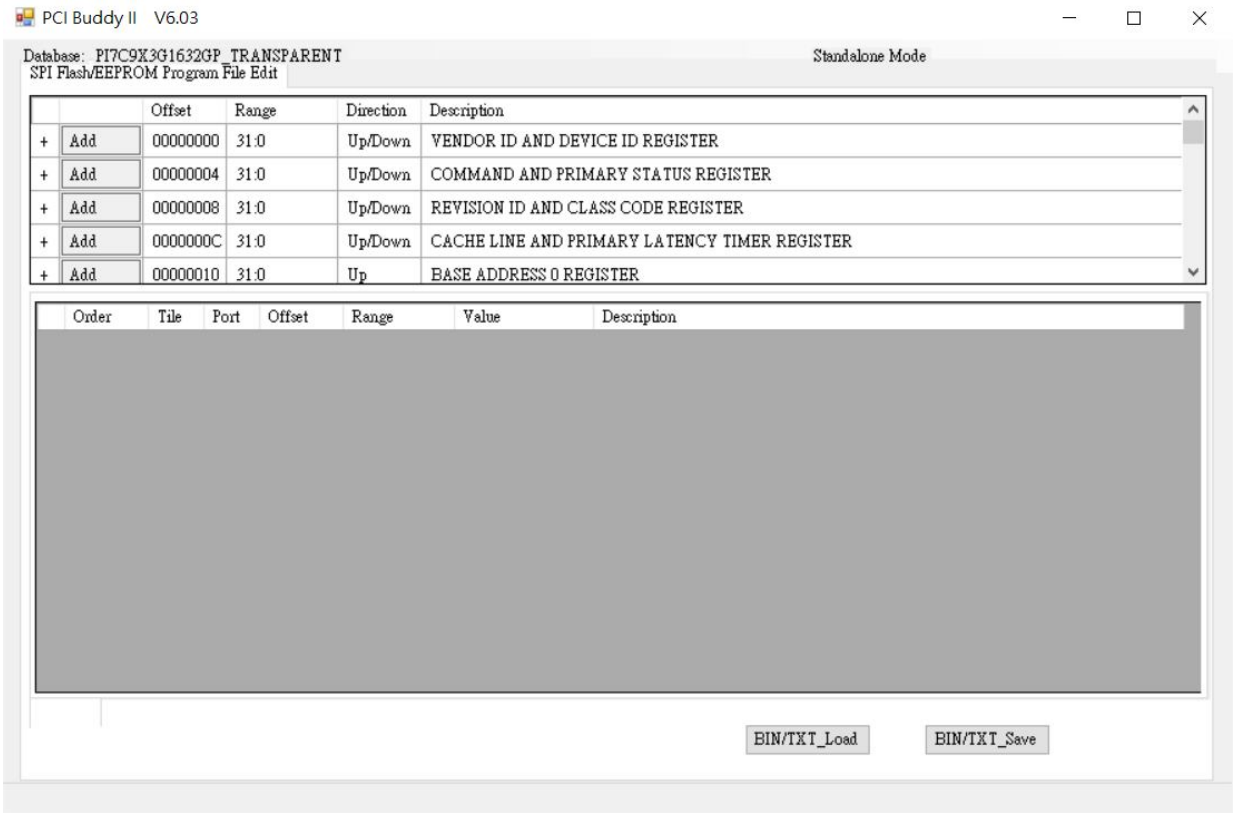


Out-Band Mode：



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Standalone Mode：



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4 Usage

4.1 Selection which mode want to perform

Executing PCIBUDDY_II.exe, it will search all Diodes GEN3 switch products in system.

At In-Band mode, PCI-BUDDY would check all Diodes Gen3 switch ports, and display to the first Tab. You can find information related to Bus#, Device#, Function#, Port#, Port Type, Device ID and description for each port in the first configuration tab. By using mouse and keyboard to select which port you want to access.

At Out-Band mode, just like In-Band mode, but you need to set related I2C address / Tile / Port for which tile/port you want to access.

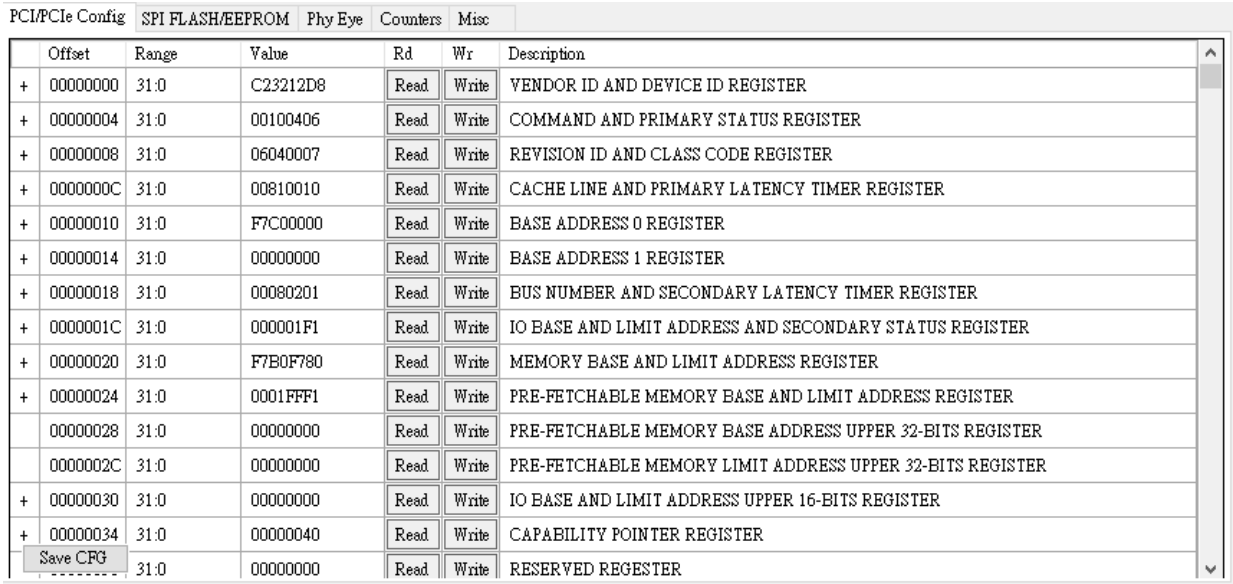
At Standalone mode, it provides you an environment to edit/create *.bin file or *.txt for EEPROM or SPI-Flash to use, no Diodes Gen3 Switch product is needed.

In other tabs, there are 「SPI Flash / EEPROM」, 「Phy Eye」, 「LTSSM」, 「Counters」 and 「Misc」 function Tabs. Among the tabs, 「LTSSM」 function is special function for Out-Band mode. User can use mouse to select which function you want to execute.

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4.2 PCIE Config Tab

Each port supports this function. It is used to read/write all configuration registers.



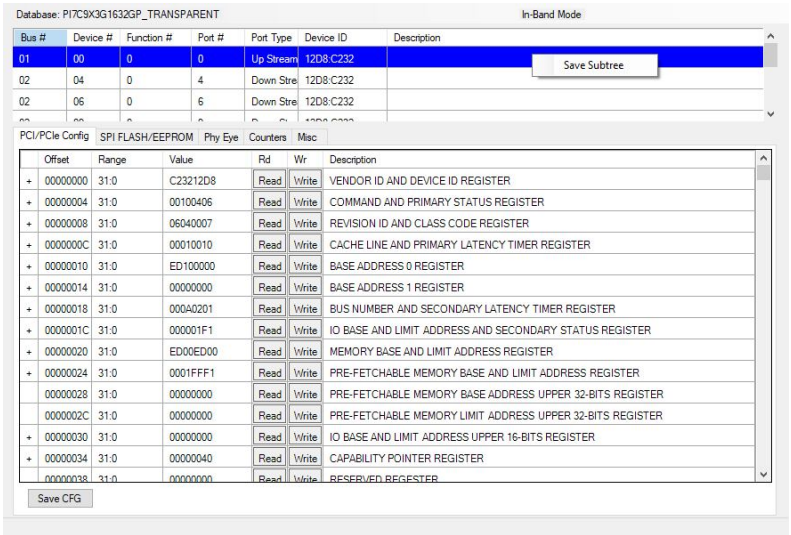
	Offset	Range	Value	Rd	Wr	Description
+	00000000	31:0	C23212D8	Read	Write	VENDOR ID AND DEVICE ID REGISTER
+	00000004	31:0	00100406	Read	Write	COMMAND AND PRIMARY STATUS REGISTER
+	00000008	31:0	06040007	Read	Write	REVISION ID AND CLASS CODE REGISTER
+	0000000C	31:0	00810010	Read	Write	CACHE LINE AND PRIMARY LATENCY TIMER REGISTER
+	00000010	31:0	F7C00000	Read	Write	BASE ADDRESS 0 REGISTER
+	00000014	31:0	00000000	Read	Write	BASE ADDRESS 1 REGISTER
+	00000018	31:0	00080201	Read	Write	BUS NUMBER AND SECONDARY LATENCY TIMER REGISTER
+	0000001C	31:0	000001F1	Read	Write	IO BASE AND LIMIT ADDRESS AND SECONDARY STATUS REGISTER
+	00000020	31:0	F7B0F780	Read	Write	MEMORY BASE AND LIMIT ADDRESS REGISTER
+	00000024	31:0	0001FFF1	Read	Write	PRE-FETCHABLE MEMORY BASE AND LIMIT ADDRESS REGISTER
	00000028	31:0	00000000	Read	Write	PRE-FETCHABLE MEMORY BASE ADDRESS UPPER 32-BITS REGISTER
	0000002C	31:0	00000000	Read	Write	PRE-FETCHABLE MEMORY LIMIT ADDRESS UPPER 32-BITS REGISTER
+	00000030	31:0	00000000	Read	Write	IO BASE AND LIMIT ADDRESS UPPER 16-BITS REGISTER
+	00000034	31:0	00000040	Read	Write	CAPABILITY POINTER REGISTER
	[illegible]	31:0	00000000	Read	Write	RESERVED REGESTER

There are 7 columns, “+/-“, “Offset”, “Range”, “Value”, “Rd”, “Wr” and “Description”.

+/-：+… only display bit[31:0], -… display each field for this register
Offset：offset value for this register
Range：bit mapping for this register
Value：current value for this register
Rd：read again after click for this register
Wr：write after click (Value column should be changed and then click Wr Tab)
Description：register name. Double click “Description”, and the datasheet will be open to corresponding page [Note]

[Note]：This feature of datasheet link needs Adobe’s “AcroRd32.exe” support. Some versions of Windows OS can’t fully support this feature yet. Please confirm the Adobe Reader 11 has been installed and AcroRd32.exe is on running folder.

Save Subtree feature (In-band mode only):
When press right button of mouse, show out the “Save Subtree”. It can able to save all cfg registers from RC to EP devices.

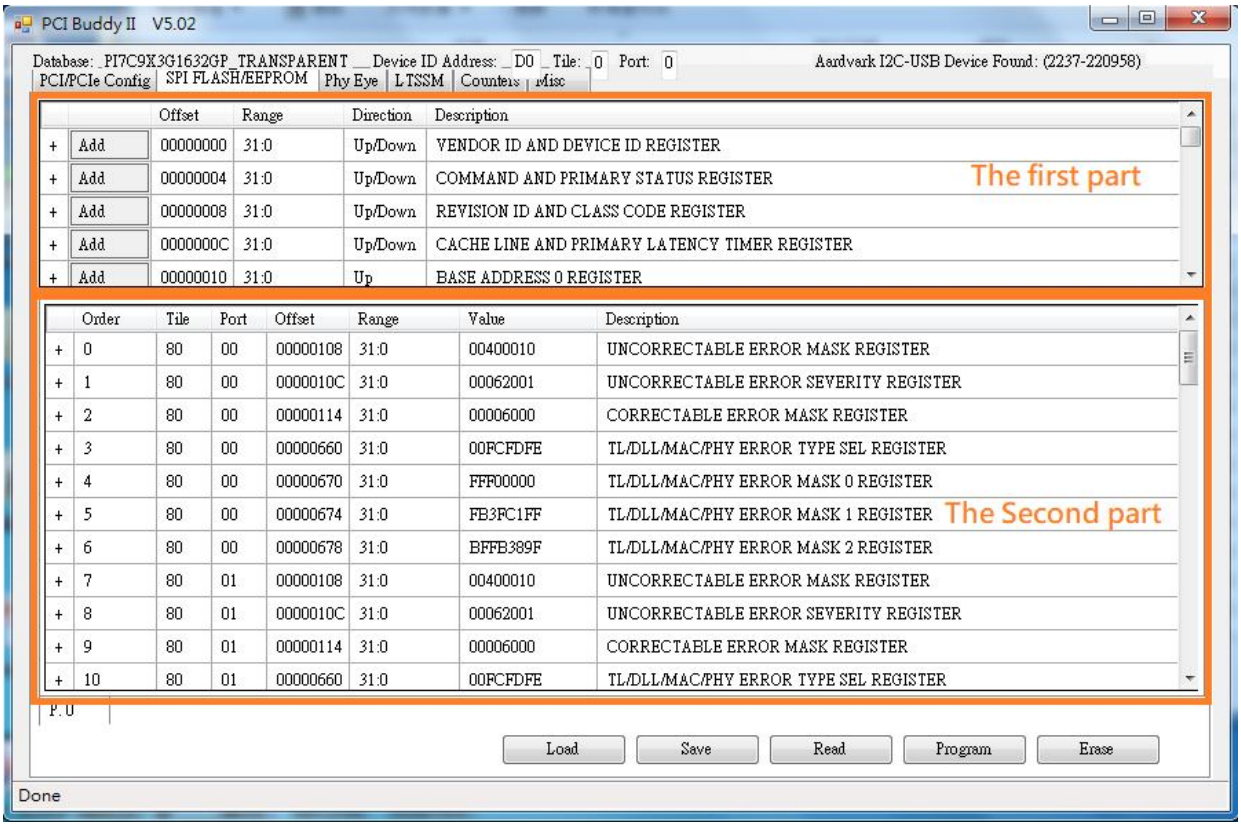


	Offset	Range	Value	Rd	Wr	Description
+	00000000	31:0	C23212D8	Read	Write	VENDOR ID AND DEVICE ID REGISTER
+	00000004	31:0	00100406	Read	Write	COMMAND AND PRIMARY STATUS REGISTER
+	00000008	31:0	06040007	Read	Write	REVISION ID AND CLASS CODE REGISTER
+	0000000C	31:0	00010010	Read	Write	CACHE LINE AND PRIMARY LATENCY TIMER REGISTER
+	00000010	31:0	ED100000	Read	Write	BASE ADDRESS 0 REGISTER
+	00000014	31:0	00000000	Read	Write	BASE ADDRESS 1 REGISTER
+	00000018	31:0	000A0201	Read	Write	BUS NUMBER AND SECONDARY LATENCY TIMER REGISTER
+	0000001C	31:0	000001F1	Read	Write	IO BASE AND LIMIT ADDRESS AND SECONDARY STATUS REGISTER
+	00000020	31:0	ED00ED00	Read	Write	MEMORY BASE AND LIMIT ADDRESS REGISTER
+	00000024	31:0	0001FFF1	Read	Write	PRE-FETCHABLE MEMORY BASE AND LIMIT ADDRESS REGISTER
	00000028	31:0	00000000	Read	Write	PRE-FETCHABLE MEMORY BASE ADDRESS UPPER 32-BITS REGISTER
	0000002C	31:0	00000000	Read	Write	PRE-FETCHABLE MEMORY LIMIT ADDRESS UPPER 32-BITS REGISTER
+	00000030	31:0	00000000	Read	Write	IO BASE AND LIMIT ADDRESS UPPER 16-BITS REGISTER
+	00000034	31:0	00000040	Read	Write	CAPABILITY POINTER REGISTER
	00000038	31:0	00000000	Read	Write	RESERVED REGESTER

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4.3 SPI Flash / EEPROM Tab

Only upstream (P0) port supports this function. It is used to read/write EEPROM. Following is screenshot from Out-Band mode



There are two parts in the entire window, the top of the window shows the current port you want to see, the first part displays all CFG registers, and the second part display EEPROM content.

At the first part, user can add register to the second part by clicking “Add” tab.

At the second part, there are eight columns, 「+/-」, 「Order」, 「Tile」, 「Port」, 「Offset」, 「Range」, 「Value」 and 「Description」.

- +/-：+… DW display bit[31:0], -… display each field for this register
- Order：sequence number for this register
- Tile：tile number for this register
- Port：port number for this register
- Offset：offset for this register
- Range：bit mapping
- Value：the value for this register
- Description：the register name. Double click “Description”, the datasheet will be open to corresponding page [Note]

[Note]：This feature of datasheet link needs Adobe’s “AcroRd32.exe” support. Some versions of Windows OS can’t fully support this feature yet. Please confirm the Adobe Reader 11 has been installed and AcroRd32.exe is on running folder.

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At the bottom of the entire window, there are five Function Tabs, 「Load」, 「Save」, 「Read」, 「Program」, and 「Erase」.

- Load：Load EEPROM content from file and display to the third part
- Save：Save EEPROM content in the third part to a file
- Read：Read EEPROM by CFG registers and display to the third part
- Program：Program EEPROM content in the third part to EEPROM
- Erase：Erase EEPROM. (Warning: This process will lose entire data in EEPROM！)

User can modify register directly in field value by clicking the left key of mouse to edit the value, and then double click to confirm. By pressing the left key of Mouse and move UP/DN can change 「Order」 of the registers.
If you want to delete the register you don't want, clicking the right key of Mouse.

For EEPROM exported content：

The contents of EEPROM for switch partition mode are defined as below. The EEPROM is divided into several sections (according to tile number) and each section is corresponding to one tile. The starting EEPROM address for each tile is shown in the following table.

EEPROM Address	Value	Description
TILE0		
00h	1516h	EEPROM signature
02h	EEPROM_BYTE_SIZE	EEPROM size byte count
04h	CFG_Tile_Port	1st Configuration tile_port Address Bit[7:0]: port number Bit[15:8]: tile number
06h	CFG_OFFSET_ADDR	1st Configuration Register Address
08h	CFG_LOW_DATA	1st Configuration Register Data (low word)
0Ah	CFG_HIGH_DATA	1st Configuration Register Data (high word)
…	…	…
TILE1		
2aa0h	1516h	EEPROM signature
…	…	…

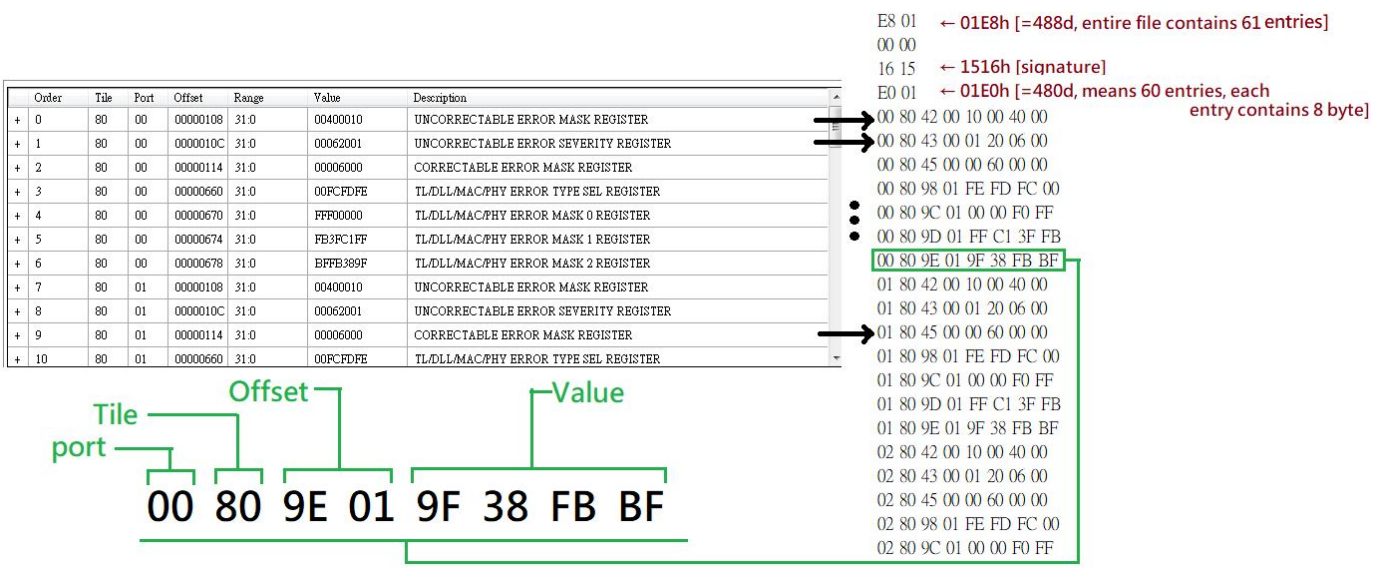
To compare the PCIBUDDY interface and EEPROM exported content：

A. The first word must contain the signature (i.e. 1516h). If not, EEPROM will stop loading value into configuration registers of packet switch

B. The second word indicates the byte count for all entries that are going to pre-set a desired value into configuration registers. One entry represents 8 bytes. For example, if there are four entries stored in EEPROM, the EEPROM_BYTE_SIZE is 32.

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Following take 3G1632GP EEPROM for example：



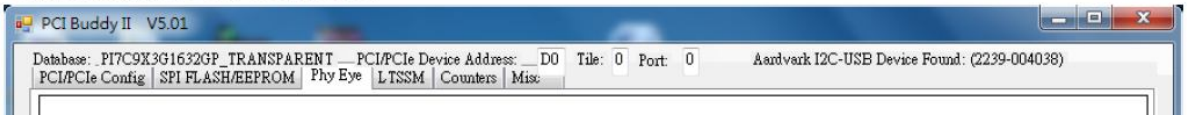
	Order	Tile	Port	Offset	Range	Value	Description
+	0	80	00	00000108	31:0	00400010	UNCORRECTABLE ERROR MASK REGISTER
+	1	80	00	0000010C	31:0	00062001	UNCORRECTABLE ERROR SEVERITY REGISTER
+	2	80	00	00000114	31:0	00006000	CORRECTABLE ERROR MASK REGISTER
+	3	80	00	00000660	31:0	00FCFDFE	TL/DLL/MAC/PHY ERROR TYPE SEL REGISTER
+	4	80	00	00000670	31:0	FFF00000	TL/DLL/MAC/PHY ERROR MASK 0 REGISTER
+	5	80	00	00000674	31:0	FB3FC1FF	TL/DLL/MAC/PHY ERROR MASK 1 REGISTER
+	6	80	00	00000678	31:0	BFFB389F	TL/DLL/MAC/PHY ERROR MASK 2 REGISTER
+	7	80	01	00000108	31:0	00400010	UNCORRECTABLE ERROR MASK REGISTER
+	8	80	01	0000010C	31:0	00062001	UNCORRECTABLE ERROR SEVERITY REGISTER
+	9	80	01	00000114	31:0	00006000	CORRECTABLE ERROR MASK REGISTER
+	10	80	01	00000660	31:0	00FCFDFE	TL/DLL/MAC/PHY ERROR TYPE SEL REGISTER

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4.4 Phy Eye Tab

PHY eye function are implemented to Port 0 only (Tile0 P0 or Tile1 P0)
Setting step：

1) Set Port at Port 0 at the top.



2) Select Phy and Lane numbers first. After set Phy and Lane numbers



3) Click 「Capture」 Tab, PCIBUDDY will start to draw RX eye

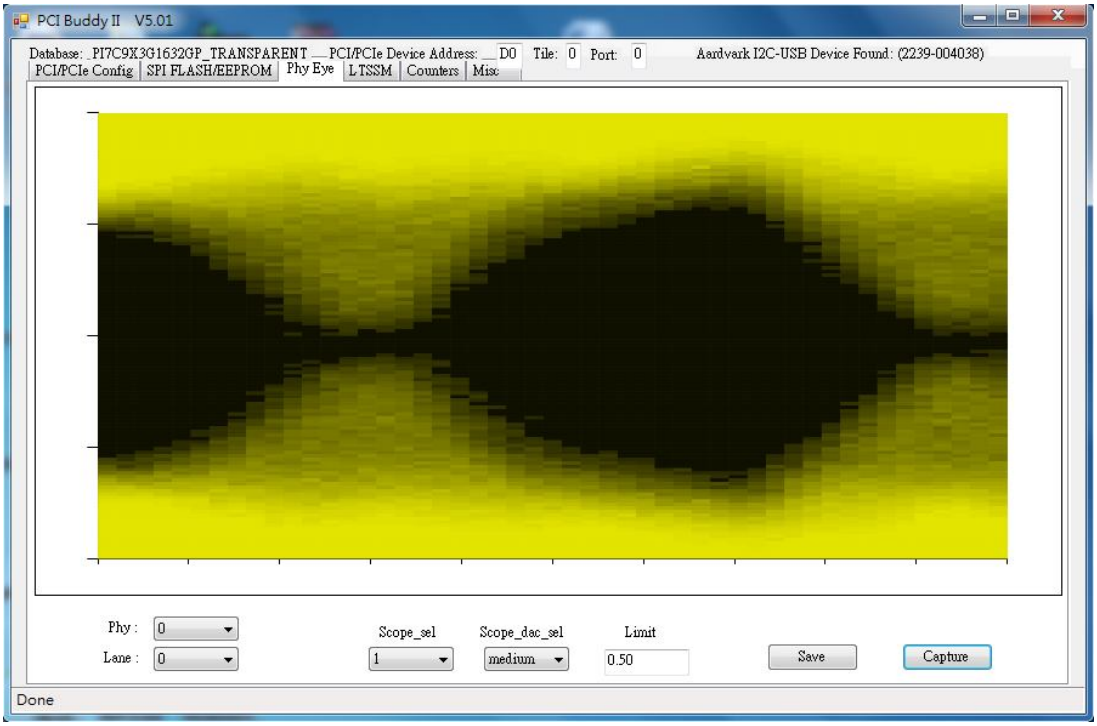
Before 「Done」 message is shown in the bottom line, please do not click any function tabs.

Following list Gen3 Packet Switch Port-Lane Mapping. Each tile includes 4 QPHY, and each QPHY includes four lanes. The mapping table is as below.

PHY#	Lane	606GP			808GP					816GP (1632GP = 2 Tile 816GP)				
		306	506	606	208	308	408	508	808	216	316	416	516	816
0	Lane 0	P0	P0	P0	P0	P0	P0	P0	P0	P0	P0	P0	P0	P0
0	Lane 1	P0	P0	P1	P0	P0	P0	P0	P1	P0	P0	P0	P0	P0
0	Lane 2				P0	P0	P1	P0	P2	P0	P0	P0	P0	P1
0	Lane 3				P0	P0	P1	P0	P3	P0	P0	P0	P0	P1
1	Lane 0									P0	P0	P2	P0	P2
1	Lane 1									P0	P0	P2	P0	P2
1	Lane 2									P0	P0	P2	P0	P3
1	Lane 3									P0	P0	P2	P0	P3
2	Lane 0	P4	P4	P4	P4	P4	P4	P4	P4	P4	P4	P4	P4	P4
2	Lane 1	P4	P5	P5	P4	P4	P4	P5	P5	P4	P4	P4	P4	P4
2	Lane 2	P5	P6	P6	P4	P5	P5	P6	P6	P4	P4	P4	P5	P5
2	Lane 3	P5	P7	P7	P4	P5	P5	P7	P7	P4	P4	P4	P5	P5
3	Lane 0									P4	P6	P6	P6	P6
3	Lane 1									P4	P6	P6	P6	P6
3	Lane 2									P4	P6	P6	P7	P7
3	Lane 3									P4	P6	P6	P7	P7

Following is screenshot from Out-Band mode for 3G1632GP

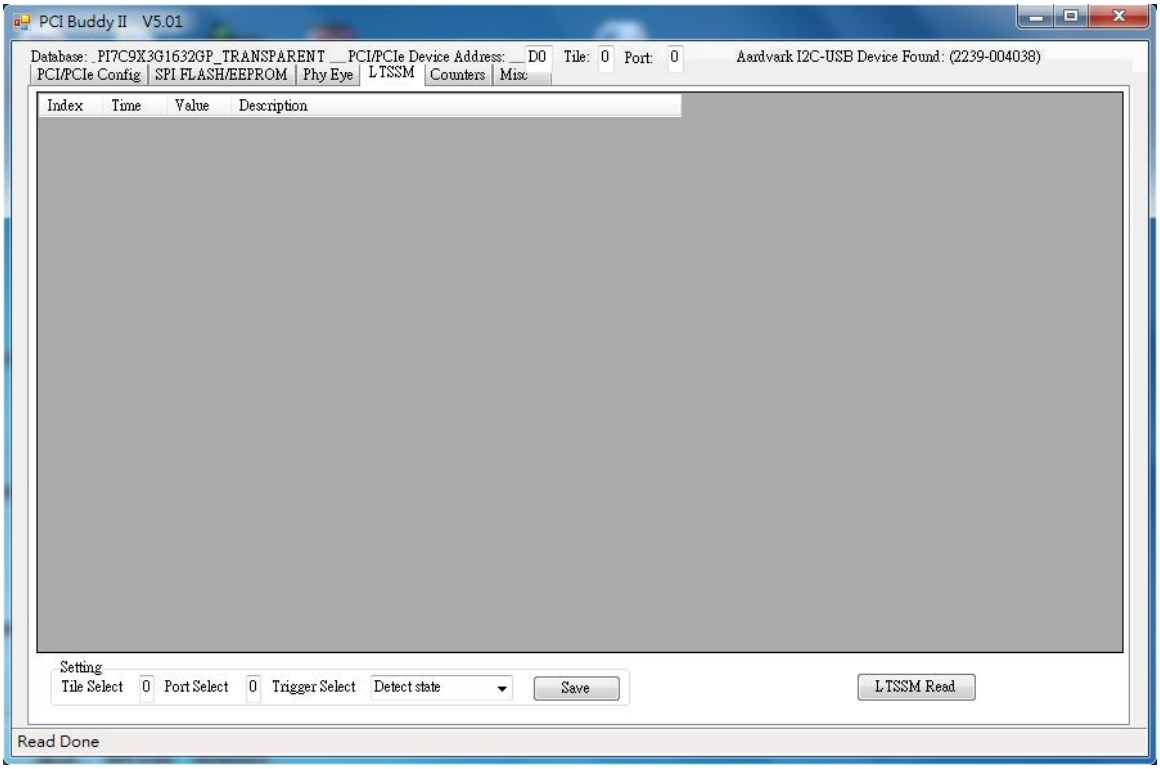
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4.5 LTSSM Tab (Only support in Out-Band mode)

Only Port 0 (P0 or P10) supports this function. It is used to display the flow of LTSSM state machines. As LTSSM occurs at power up, it need set LTSSM trigger condition by EEPROM first, and reboot system. Then, the entire LTSSM flow will be recorded to internal debug buffer, and can be read by this function.



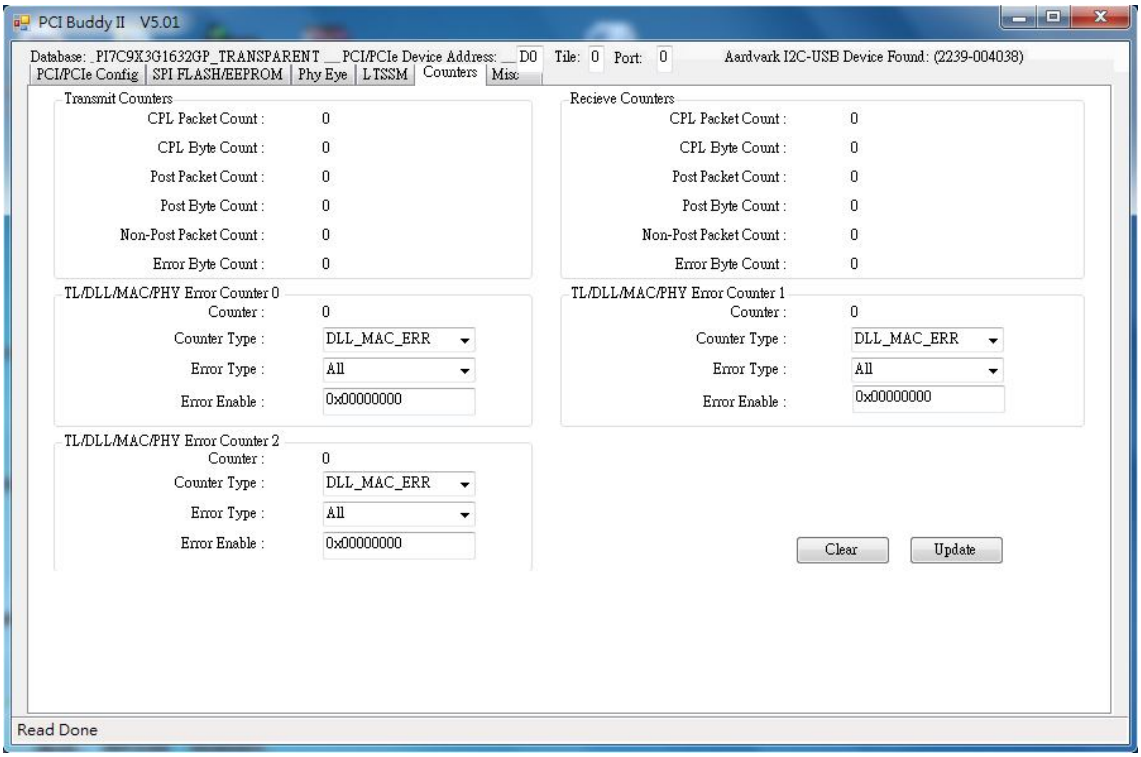
At Setting Window, select which port and what is the LTSSM trigger condition. Then, click 「Save」 Tab to save the setting to EEPROM. Before 「DONE」 message is shown at the bottom line, DO NOT click mouse. After 「DONE」 message is shown, reboot the system which PI7C9X3G1632GP plugs in.

After power up the system, clicking 「LTSSM Read」 tab, and PCIBUDDY will display the entire LTSSM flow. If want to save the LTSSM flow to a file, clicking the right key of mouse.

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4.6 Counters Tab

Each port supports this function. It is used to display transmit, receive and error counters.



At the Transmit and Receive Counters Window, it will display CPL Packet, CPL Byte, Post Packet, Post Byte, Non-Post Packet and Error Byte Counts.

At the TL/DLL/MAC/PHY Error Counter 0/1/2, it will display the error counter defined by “Counter Type”, “Error Type” and “Error Enable”. “Error Enable” setting is used if “Error Type” is set ALL. Please see datasheet to get more detail information for error counters.

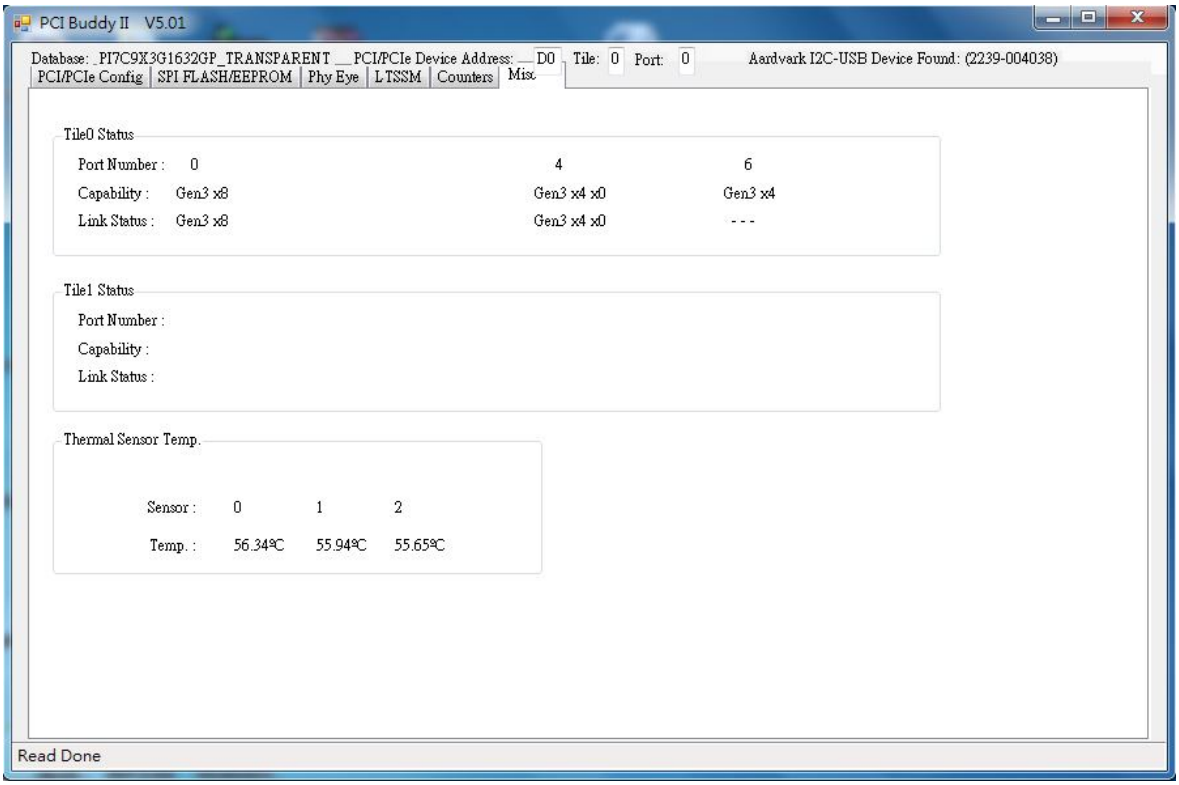
There are two Function Tabs, 「Clear」 and 「Update」

- Clear：all counters will be clear after click
- Update：all counters will be updated after click

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4.7 Misc. Tab

There are two Windows, “Tile Status” and “Thermal Sensor Temp.”. Only Port 0 (P0 or P10) supports two Windows, but left ports only support “Tile Status” Window.



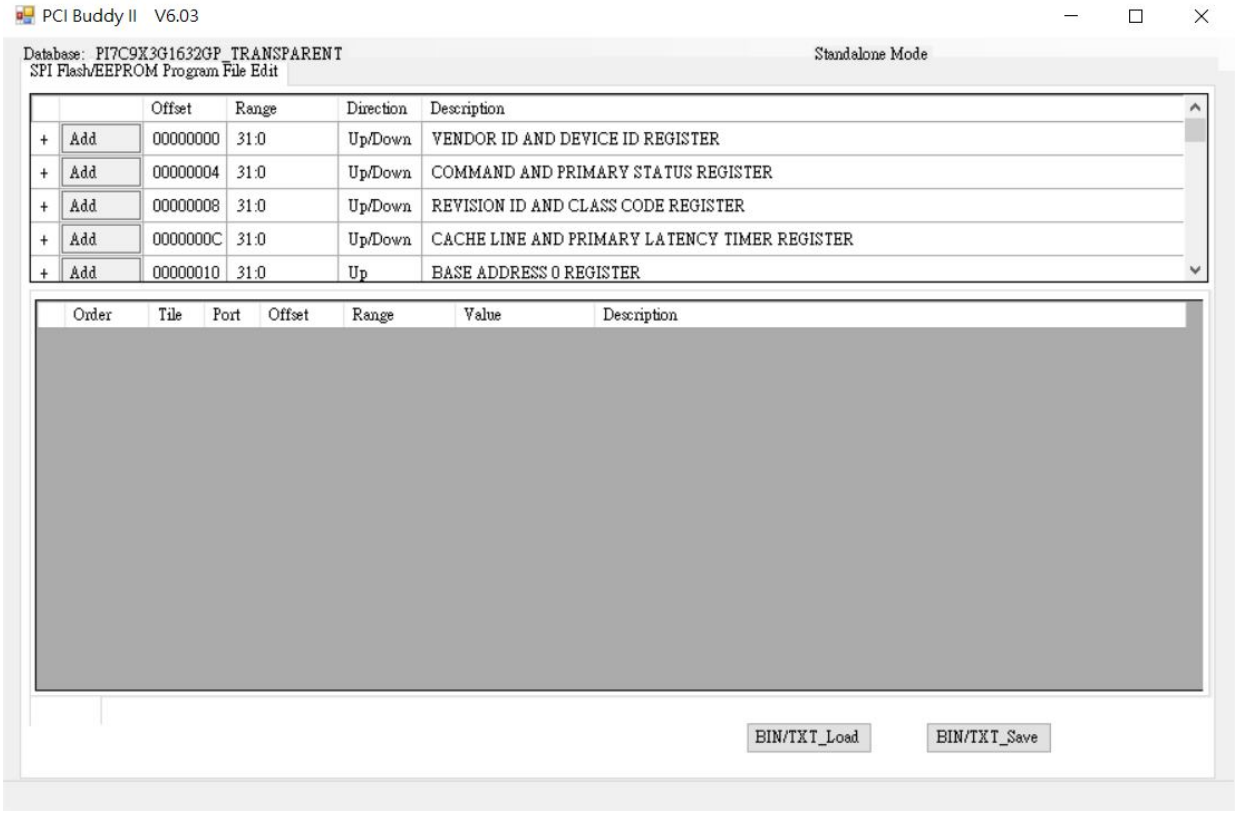
At Tile Status Window, it displays the port number, link speed and link width.

At Thermal Sensor Temp. Window, it displays temperatures for three thermal sensors.

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4.8 SPI-Flash/EEPROM Program File Edit

Same interface as 4.3 SPI-Flash/EEPROM Tab



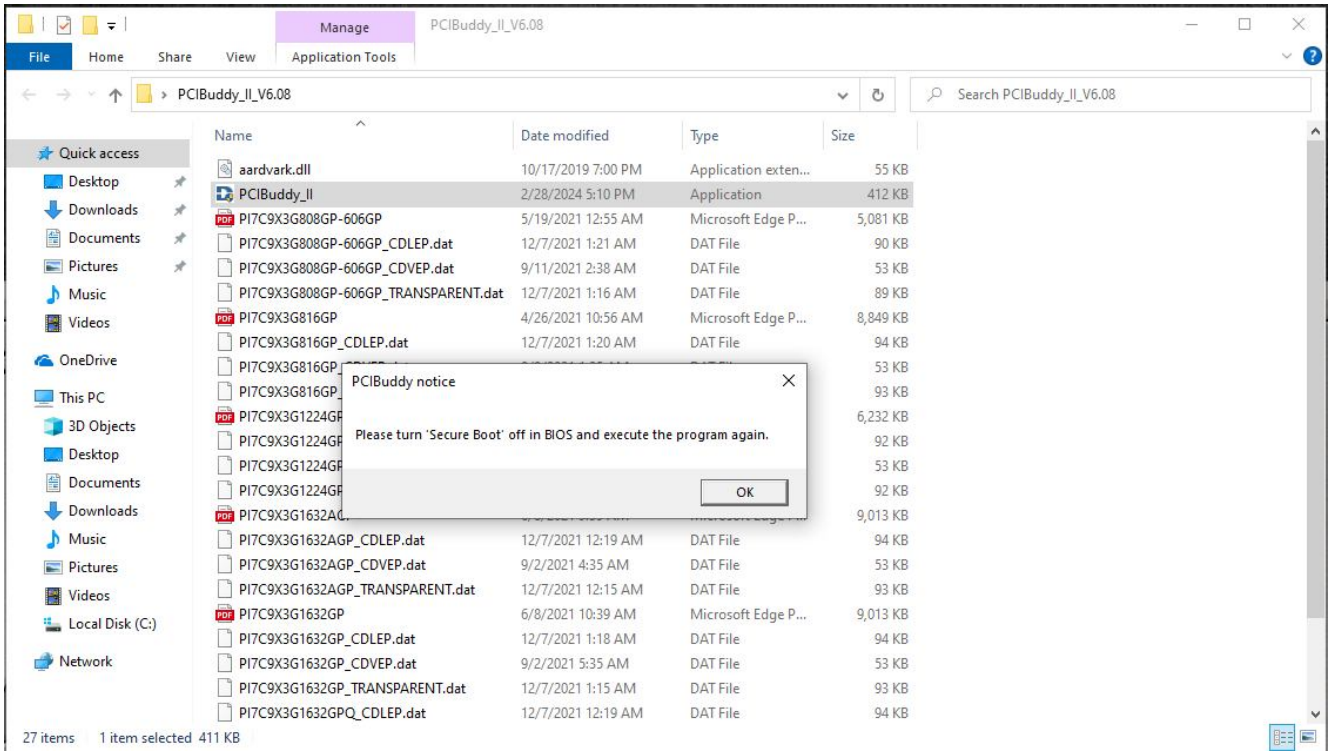
Find the register you need, by pressing 「Add」, you can add the register to the following block.
If you want to modify your existing file, just click 「BIN/TXT_Load」 and choose your file, modify it, and click 「BIN_TXT_Save」 to save the content depending on the file type you need (*.bin or *.txt)

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5 Trouble Shooting

5.1 Some old Aardvark dongles need to have the firmware updated to v3.51 or later.

5.2 PCIBUDDY tool needs to disable Secure Boot option in BIOS since v6.08 released. Or the silicon will be detecting issue under Secure Boot enable with older PCIBUDDY version.



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